THE MASTERMIND COACHING CLASSES

CBSE / RBSE · Nursery – Class XII · Road No. 5, VKIA, Jaipur · 9887258679

CLASS IX — MATHEMATICS · FULL SYLLABUS EXAMINATION

Session 2026-27 | CBSE / RBSE Pattern

Class: 9	Subject: Maths	Set 02	Time: 3 Hours	Max. Marks: 80

General Instructions:

1. This question paper contains 38 questions and is divided into 5 sections — A, B, C, D and E.
2. In Section A, Q. 1–18 are multiple-choice questions and Q. 19–20 are Assertion–Reason based questions of 1 mark each.
3. In Section B, Q. 21–25 are very short answer (VSA) type questions carrying 2 marks each.
4. In Section C, Q. 26–31 are short answer (SA) type questions carrying 3 marks each.
5. In Section D, Q. 32–35 are long answer (LA) type questions carrying 5 marks each.
6. In Section E, Q. 36–38 are case-study based questions carrying 4 marks each with sub-parts of 1, 1 and 2 marks respectively.
7. All questions are compulsory. However, an internal choice is provided in 1 question of Section B, 2 questions of Section C and 2 questions of Section D. An internal choice is provided in all the 2-mark sub-parts of Section E.
8. Draw neat and clean figures wherever required. Take π = 22/7 wherever required, if not stated. Use of calculators is not allowed.
9. Write the serial number of each question before attempting it.

SECTION A — Objective Type — 20 questions · 20 marks

Q1–Q18 are multiple-choice questions (1 mark each). Q19–Q20 are Assertion (A) and Reason (R) questions: select the correct option from (a), (b), (c), (d) given below each.

1. A point P(a, b) lies in the second quadrant. Then: **1**

(a) $a < 0$ and $b > 0$ (b) $a > 0$ and $b > 0$

(c) $a < 0$ and $b < 0$ (d) $a > 0$ and $b < 0$

2. The distance between the points (−2, 5) and (−2, −3) is: **1**

(a) 8 units (b) 5 units (c) √8 units (d) 2 units

3. The reflection of the point (4, −7) in the y-axis is: **1**

(a) (−4, 7) (b) (4, 7) (c) (4, −7) (d) (−4, −7)

4. Which of the following is a linear polynomial? **1**

(a) $\sqrt{x} + 3$ (b) $x^2 - 1$ (c) $2x - 5$ (d) $1/x + 2$

5. The zero of the polynomial $p(x) = 5x - 15$ is: **1**

(a) 3 (b) −3 (c) 5 (d) 15

6. A notebook costs ₹x and a pen costs ₹20. The cost (in ₹) of 6 notebooks and one pen is: **1**

(a) 26x (b) 6x − 20 (c) x + 120 (d) 6x + 20

7. Which of the following numbers lies between √2 and √3? **1**

(a) 1.3 (b) 2 (c) 1.5 (d) 1.8

8. The decimal expansion of √5 is: **1**

(a) terminating (b) 1.5

(c) non-terminating and non-repeating (d) non-terminating and repeating

9. If $|x| = \frac{5}{3}$ for a rational number x, then x is: **1**

(a) ± 5/3 (b) 0 (c) −5/3 only (d) 5/3 only

10. On simplification, $(x + 4)(x - 4)$ is equal to: **1**

(a) $x^2 + 8x - 16$ (b) $x^2 + 16$ (c) $x^2 - 16$ (d) $x^2 - 8x + 16$

11. Using a suitable identity, 51^2 is equal to: **1**

(a) 2651 (b) 2601 (c) 2501 (d) 2611

12. Two equal chords AB and CD of a circle with centre O subtend ∠AOB = 70° at the centre. Then ∠COD is: **1**

(a) 110° (b) 140° (c) 70° (d) 35°

13. The longest chord of a circle of diameter 18 cm has length: **1**

(a) 36 cm (b) √18 cm (c) 9 cm (d) 18 cm

14. An equilateral triangle has perimeter 36 cm. Its area (by Heron's formula) is: **1**

(a) $36\sqrt{3}$ cm² (b) $72\sqrt{3}$ cm² (c) $18\sqrt{3}$ cm² (d) 36 cm²

15. The sides of a triangular plot are in the ratio 5 : 12 : 13 and its perimeter is 120 m. The longest side is: **1**

(a) 60 m (b) 20 m (c) 52 m (d) 48 m

16. Two coins are tossed together. The probability of getting exactly one head is: **1**

(a) 1 (b) $\frac{3}{4}$ (c) $\frac{1}{2}$ (d) $\frac{1}{4}$

17. A letter is picked at random from the word 'PROBABILITY'. The probability of picking the letter B is: **1**

(a) $\frac{4}{11}$ (b) $\frac{2}{11}$ (c) $\frac{1}{11}$ (d) $\frac{3}{11}$

18. The 20th term of the sequence 3, 7, 11, 15, … is: **1**

(a) 79 (b) 83 (c) 80 (d) 77

19. **Assertion (A):** The perpendicular drawn from the centre of a circle to a chord bisects the chord. **1**
Reason (R): The perpendicular bisector of a chord of a circle passes through the centre of the circle.

(a) Both A and R are true, and R is the correct explanation of A.
(b) Both A and R are true, but R is not the correct explanation of A.
(c) A is true but R is false.
(d) A is false but R is true.

20. **Assertion (A):** The probability of an event can be 1.2. **1**
Reason (R): The probability of any event lies between 0 and 1, both inclusive.

(a) Both A and R are true, and R is the correct explanation of A.
(b) Both A and R are true, but R is not the correct explanation of A.
(c) A is true but R is false.
(d) A is false but R is true.

SECTION B — Very Short Answer (2 marks each) — 5 questions · 10 marks

Internal choice is provided in one question.

21. The side of a square is (3x − 2) cm. Write the expression for its perimeter and find its value for x = 4. **2**

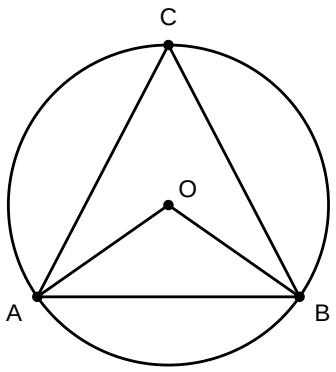
22. Without actual division, state which of $\frac{13}{40}$ and $\frac{7}{30}$ has a terminating decimal expansion. Give a reason. **2**

OR

Find two rational numbers between $\frac{2}{3}$ and $\frac{3}{4}$.

23. Evaluate 103 × 107 using a suitable identity. **2**

24. In the figure, O is the centre of the circle and ∠ACB = 62°. Find ∠OAB. **2**



25. Find the second, third and fourth terms of the sequence given by $t_1 = 2$ and $t_n = (t_{n-1})^2 - 1$ for $n \geq 2$. **2**

SECTION C — Short Answer (3 marks each) — 6 questions · 18 marks

Internal choice is provided in two questions.

26. Show that $0.428571428571\ldots = \frac{3}{7}$. **3**

27. In a school, 100 students chose their favourite sport: **3**

Sport	Cricket	Football	Kabaddi	Hockey
Students	40	25	20	15

A student is chosen at random. Find the probability that the student chose (i) kabaddi, (ii) cricket or hockey, (iii) not football.

OR

The ages of 100 members of a club were recorded:

Age (years)	11	12	13	14	15
Members	12	18	30	25	15

A member is chosen at random. Find the probability that the member is (i) 13 years old, (ii) older than 13 years, (iii) younger than 12 years.

28. A(−3, 2), B(5, 2) and C(5, −4) are three vertices of a rectangle ABCD. Find (i) the coordinates of D, (ii) the length of the diagonal AC, (iii) the area of the rectangle. **3**

29. Expand: $(2x - 3y + 1)^2$. **3**

30. In a circle with centre O, chords AB and CD are equal. Prove that they subtend equal angles at the centre. **3**

31. Find the area of the triangle whose sides are 7 cm, 15 cm and 20 cm, using Heron's formula. **3**

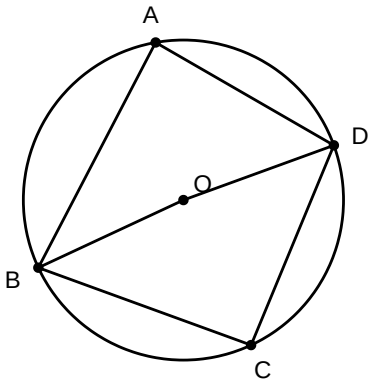
OR

Using Heron's formula, find the area of an equilateral triangle of side 16 cm. (Leave the answer in terms of √3.)

SECTION D — Long Answer (5 marks each) — 4 questions · 20 marks

Internal choice is provided in two questions.

32. Prove that the sum of either pair of opposite angles of a cyclic quadrilateral is 180°. **5**



33. (i) Express 3.2444… (that is, 3.24 with the 4 repeating) in the form p/q in lowest terms. (2½) **5**

(ii) If $x = -\frac{7}{4}$ and $y = \frac{5}{6}$, find $|x - y|$ and $|x| \cdot |y|$. (2½)

34. How many two-digit numbers are divisible by 4? Find their sum. (2½ + 2½) **5**

OR

The rungs of a ladder decrease uniformly in length from 45 cm at the bottom to 25 cm at the top. The rungs are 25 cm apart and the top rung is 2½ m above the bottom rung. Find the number of rungs and the total length of wood needed for the rungs.

35. A triangular park has sides 15 m, 20 m and 25 m. Find (i) its area using Heron's formula, (ii) the cost of fencing it at ₹30 per metre, (iii) the cost of turfing it at ₹8 per m². (2 + 1½ + 1½) **5**

OR

An isosceles triangle has perimeter 36 cm and each of its equal sides is 13 cm. Find the area of the triangle and the length of the altitude on its base.

SECTION E — Case-Study Based Questions (4 marks each) — 3 questions · 12 marks

Read each case carefully and answer the questions that follow. Internal choice is provided in every 2-mark sub-part.

36. Aarav starts saving for a school trip. He saves ₹100 in the first month and increases his saving by ₹50 every month, so his saving in the n-th month is $t_n = 50n + 50$.

(i) How much does he save in the 6th month? 1

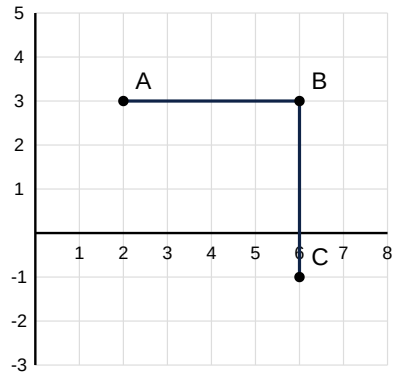
(ii) The monthly savings form an AP. Write its common difference. 1

(iii) In which month will he save ₹1000? 2

OR

(iii) Find the total amount he saves in the first 10 months.

37. The map of a colony is drawn on a grid (1 unit = 100 m). Meera's house is at A(2, 3), Karan's house is at B(6, 3) and Zoya's house is at C(6, −1).



(i) Find the distance between Meera's and Karan's houses (in units). 1

(ii) The school is at the midpoint of AC. Find its coordinates. 1

(iii) A park is to be located at D so that ABCD is a square. Find the coordinates of D. 2

OR

(iii) Find the straight-line distance between the houses of Meera and Zoya.

38. A spinner is divided into 8 equal sectors numbered 1 to 8. The spinner is spun once and the number on the sector that stops at the pointer is noted.

(i) Find the probability of getting the number 7. 1

(ii) Find the probability of getting a prime number. 1

(iii) Find the probability of getting a number greater than 5. 2

OR

(iii) Find the probability of getting a multiple of 3 or a multiple of 4.